

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

EE2102-1 – ELECTRICAL MACHINES-I (E&E)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- For transformer given, turns ratio is equal to a , what will be the impedance of primary with respect to secondary?
 - a^2 times the secondary impedance
 - ' a ' times secondary impedance
 - secondary impedance / a
 - secondary impedance / a^2
- The SC current required for testing a 230/230V, 2kVA transformer is
 - 8.69A
 - 6.14A
 - 4.34A
 - 3.07A
- A transformer can have zero regulation at
 - Zero power factor
 - Unity power factor
 - Leading power factor
 - Lagging power factor
- The full load copper loss of a transformer is 1600W, its value at half full load is
 - 800W
 - 400W
 - 1131W
 - 1000W
- The primary and secondary induced EMFs, E_1 and E_2 in a two winding transformer are always
 - Antiphase with each other
 - In-phase with each other
 - Equal in magnitude
 - Determined by load on secondary
- Which of the following is the wrong expression?
 - $i_1 N_1 = i_2 N_2$
 - $i_1 V_1 = i_2 V_2$
 - $i_1 N_2 = i_2 N_1$
 - $V_2 N_1 = V_1 N_2$
- The increase in kVA capacity produced by connecting a 2 winding transformer as an auto-transformer is due to
 - increase in turn ratio
 - increase in secondary voltage
 - increase in transformer efficiency
 - establishment of conductive link between primary and secondary.
- In three phase bank phases are _____
 - the phases are electrically not connected and magnetically independent
 - the phases are electrically connected and magnetically independent
 - the phases are electrically not connected and magnetically dependent
 - the phases are electrically connected and magnetically dependent
- In an induction motor, when the number of stator slots is equal to an integral number of rotor slots
 - There may be a discontinuity in torque slip characteristics
 - The maximum torque will be high
 - A high starting torque will be available
 - The machine may fail to start
- A 3-phase induction motor runs at almost 1000 rpm at no load and 950 rpm at full load when supplied with power from a 50 Hz, 3-phase supply. The corresponding speed of the rotor field with respect to the rotor is
 - 30 revolution per minute
 - 60 revolution per minute
 - 50 revolution per minute
 - 40 revolution per minute

11. From the no load test performed on the 3-4 induction motor, following parameters can be determined.
- A) Rotor resistance and rotor reactance B) Rotor resistance
C) Sum of leakage and magnetizing reactance ($x_1 + x_m$) and constant losses D) Rotor reactance
12. A 10kW, 50 Hz, 3-phase induction motor develops the rated torque at 1440rpm. If the load torque is reduced to half, then the motor speed is?
- A) 1470rpm B) 1400rpm
C) 1410rpm D) 1444rpm
13. During starting, the external rotor resistance in slip ring induction motor is _____
- A) gradually cut out as motor picks up the speed B) fixed for a stable point
C) removed instantly as the motor is started D) added or reduced as per the efficiency requirement
14. Induction machine can be operated in the generator mode by,
- A) Increasing the speed above synchronous speed B) Increasing the load torque
C) Change in the phase sequence of supply voltage D) Increasing the rotor resistance
15. At standstill the equivalent of the 3-phase induction motor is similar to _____
- A) short-circuited two winding transformer B) open circuit two winding transformer
C) stand still synchronous motor D) Short-circuited autotransformer
16. Induction generators will deliver power at _____
- A) lagging p.f. B) leading p.f.
C) Zero p.f. D) Unity p.f.
17. Changing the rotor resistance will also affect _____
- A) starting torque, speed, current B) maximum torque, speed, current, starting torque
C) maximum torque, speed, current D) speed and current
18. In double cage induction motor, the inner cage has
- A) High resistance and high leakage reactance B) Low resistance and high leakage reactance
C) High resistance and low leakage reactance D) Low resistance and low leakage reactance
19. The starting winding of a single-phase motor is placed in ?
- A) armature B) field
C) Stator D) rotor
20. If the centrifugal switch does not open at 70 to 80 percent of synchronous speed of motor, it would result in?
- A) Damage to the starting winding B) damage to the centrifugal switch
C) Overloading of running winding D) none of the above

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Derive the emf equation of a transformer. Develop the phasor diagram and equivalent circuit of the transformer on no-load. Explain the components of equivalent circuit.	08	L*1	2	1,2
	b) A 400/100V, 5kVA, 50Hz, two winding single-phase transformer is to be used as an autotransformer to work to supply power at 400V from 500V source. Draw the connection diagram and determine the kVA rating of the autotransformer.	08	L3	2	1,2
2.	a) Discuss regenerative test on transformer with required connection diagram. What are the outcomes of this test? How can it be used for measurement of efficiency	08	L2	1	1,2
	b) 5,000-kVA, 3-phase transformer, 6.6/33-kV, Δ/Y , has a no-load loss of 15 kW and a full-load loss of 50 kW. The impedance drop at full-load is 7%. Calculate the primary voltage when a load of 3,200 kW at 0.8 p.f. is delivered at 33 kV.	08	L3	2	1,2
3.	a) Two 2,200/110V, transformers are operated in parallel to share a load of 125 kVA at 0.8 power factor lagging. Transformers are rated as below: A: 100 kVA; 0.9% resistance and 10% reactance B: 50 kVA; 1.0% resistance and 5% reactance Find the load carried by each transformer.	08	L3	1	1,2

- b) Explain the open delta connection of two single phase transformers for three phase applications. Comment on the power supplied by the bank. List the advantages and disadvantages.

08 L2 2 1,2

Unit – II

4. a) Draw the phasor diagram of induction motor. Derive an expression for the frequency of rotor current in an induction motor. Justify the power factor of an induction motor running on load is higher compared to no load condition
- b) The power input to the rotor of 440V, 50Hz, 6-pole 3-phase induction motor is 80kW. The rotor emf is observed to make 100 complete alternations per min. Calculate (i) slip, (ii) the rotor speed (iii) mechanical power developed (iv) rotor copper loss per phase
5. a) Show that in a 3 phase Induction motor, ratio of rotor copper loss to mechanical power developed is $\left(\frac{s}{1-s}\right)$ where 's' is the slip
- b) A 100-kW (output), 3300-V, 50-Hz, 3-phase, star-connected induction motor has a synchronous speed of 500 r.p.m. The full-load slip is 1.8% and F.L. power factor 0.85. Stator copper loss = 2440 W. Iron loss = 3500 W. Rotational losses = 1200 W. Calculate (i) the rotor copper loss (ii) the line current (iii) the full-load efficiency
6. a) Explain the working principle induction motor. Explain the generation of the rotating magnetic field using analytical method
- b) Explain the working of induction generator. Why is it known as asynchronous generator? Explain the advantages and limitations of the same.

08 L2
L4 3 1,2

08 L3 4 1,2

08 L2 3 1,2

08 L3 4 1,2

08 L1 3 1,2

08 L2 4 1,2

Unit – III

7. a) It is desired to install a 3-phase cage induction motor restricting the maximum line current drawn from a 400V 3-phase supply to 120A. If the starting current is 6 times full –load current, what is the maximum permissible full-load kVA of the motor when
- It is directly connected to the mains
 - It is connected through an auto-transformer with a tapping of 60%
 - It is designed for use with star-delta starter
- b) Explain why a single phase induction motor is not self-starting with relevant diagrams and double field revolving theory
8. a) Explain deep bar and double cage induction motors and discuss how the starting torque in double cage motors is improved compared to squirrel cage motor
- b) Explain with a circuit diagram, the capacitor start single phase Induction motor. Draw and explain the torque speed characteristics

08 L3 5 2,3

08 L2 5 1,2

08 L2 5 1,2

08 L2 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
